

E-Commerce Platform Development Report

Technical Report

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Student Work Archive System
Sample Document for Testing Purposes

Abstract

This technical report documents the complete development lifecycle of a full-stack e-commerce platform. The project encompasses system design, database architecture, payment integration, and cloud deployment strategies.

The platform is built using modern web technologies including Next.js for the frontend, Node.js with Express for the backend API, and PostgreSQL for data persistence. Payment processing is handled through Stripe integration.

Key topics covered include:

- Requirements analysis and system design
- Database schema design and optimization
- RESTful API development best practices
- Payment gateway integration and security
- AWS deployment architecture
- Performance optimization techniques

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Chapter 1

Project Overview

The objective was to develop a scalable e-commerce platform capable of handling 10,000+ concurrent users. The platform supports multiple vendors, customer reviews, wishlist functionality, and various payment methods.

This section contains additional technical details and supporting information relevant to the chapter topic. The content demonstrates the depth of research and analysis conducted throughout this study.

Further considerations include implementation specifics, potential limitations, and suggestions for future improvements. These elements contribute to the comprehensive nature of this academic work.

Chapter 2

Requirements Analysis

Functional requirements were gathered through stakeholder interviews and competitor analysis. Non-functional requirements focused on performance (page load < 2s), security (PCI DSS compliance), and availability (99.9% uptime).

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Chapter 3

System Architecture

The system follows a microservices architecture with separate services for user management, product catalog, order processing, and payment handling. Services communicate through RabbitMQ message queues.

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Chapter 4

Database Design

PostgreSQL was chosen for ACID compliance and complex query support. The schema includes 25 tables with proper normalization and strategic denormalization for read performance. Redis is used for caching and session storage.

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Chapter 5

API Development

RESTful APIs were developed following OpenAPI 3.0 specification. Authentication uses JWT tokens with refresh token rotation. Rate limiting and request validation protect against abuse.

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Chapter 6

Payment Integration

Stripe integration handles credit card payments with support for multiple currencies. Webhook handlers process payment events asynchronously. PCI DSS compliance is achieved through Stripe Elements.

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Chapter 7

Deployment Strategy

The application is deployed on AWS using ECS for container orchestration, RDS for managed PostgreSQL, ElastiCache for Redis, and CloudFront for CDN. Infrastructure is managed through Terraform.

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Chapter 8

Performance Results

Load testing with 10,000 concurrent users showed average response time of 150ms. Database query optimization reduced page load time by 40%. CDN integration improved global access speeds.

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References

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